Float Power Calculation

Component	Model	Qty	Voltage (V)	Current (A)	Power per Unit (W)	Total Power (W)
Controller	ESP32 S3 DevKitC1 N16R8	1	5.0	0.500000	2.500	2.500
Pressure Sensor	Bar02	1	5.0	0.001250	0.006	0.006
ADC	ADS1115	1	5.0	0.000300	0.002	0.002
Motor	N20 12V 78RPM encoder	1	12.0	1.000000	12.000	12.000
Buck Converter	LM2596S DC-DC	1	Resistance=0.05 ohm	5.000000	1.250	1.250
Flash Memory	W25Q64	2	3.6	0.025000	0.090	0.180
Motor Driver Loss	TB6612FNG estimated loss	1	5.5	3.200000	17.600	17.600
LED Module	WS2812 5050 RGB	1	5.5	0.050000	0.275	0.275
RTC	DS3231 AT24C32	1	5.5	0.003650	0.020	0.020
Temperature Sensor	GY-BME280	1	5.0	0.001120	0.006	0.006

Display	0.96 inch I2C OLED	1	5.0	0.030000	0.150	0.150
Panel Meter	DC 0-100V 100A Dual LED meter	1	5.0	0.060000	0.300	0.300
						34.288

Total current(A) = Total power/Voltage(12V) = 34.288w/12v = **2.857A**

Ground Station

Component	Model	Qty	Voltage (V)	Current (A)	Power per Unit (W)	Total Power (W)
Controller	ESP32 S3 DevKitC1 N16R8	1	5.0	0.500000	2.500	2.500
Temperature Sensor	GY-BME280	1	5.0	0.001120	0.006	0.006
Display	0.96 inch I2C OLED	1	5.0	0.030000	0.150	0.150
						2.656

Total Current for Float main part (A) *2.857A*

Current for Ground Station (A) *0.2213A*

(Thus, we will need only a 3A fuse)